Supplementary Materials

An Analytic Hazard Kernel for Optogenetically-Driven Larval Reorientation

Table S1: Leave-One-Experiment-Out Cross-Validation Results

Leave-one-experiment-out cross-validation was performed to assess model generalization. For each of 12 experiments, the factorial NB-GLM was fit on the remaining 11 experiments and used to predict event rates for the held-out experiment.

Condition	Empirical	Predicted	Rate Ratio	Status
0→250 / Control (n=2)				
Exp 1	737	785	1.066	Pass
Exp 2	670	629	0.938	Pass
0→250 / Cycling (n=4)				
Exp 1	555	921	1.659	Fail
Exp 2	488	395	0.809	Pass
Exp 3	822	603	0.733	Fail
Exp 4	545	534	0.981	Pass
50→250 / Control (n=4)				
Exp 1	766	603	0.787	Fail
Exp 2	571	937	1.641	Fail
Exp 3	657	655	0.997	Pass
Exp 4	446	305	0.684	Fail
50→250 / Cycling (n=2)				
Exp 1	477	553	1.160	Pass
Exp 2	554	478	0.862	Pass
Summary				
Mean ± SD	—	—	1.03 ± 0.31	7/12 Pass

Table 1: Cross-validation results. Rate ratio within 0.8–1.25 indicates acceptable generalization. Pass rate = 58%.

Table S2: Model Comparison

Model	Parameters	AIC	Deviance	Notes
Fixed-effects NB-GLM	8	114,814	94,592	Primary model
NB-GLMM (1|track)	9 + 623 RE	—	—	Random intercepts

Table 2: Model comparison. The fixed-effects model was used for all reported analyses. GLMM with random track intercepts is included as a robustness check.

Table S2b: GLMM Robustness Check

A Negative Binomial GLMM with random track intercepts was fit using Bambi/PyMC to verify that the main findings are robust to hierarchical structure.

Parameter	Fixed-Effects	GLMM	Change
α (kernel amplitude)	1.005	0.971	−3.4%
α_I (intensity effect)	−0.665	−0.655	+1.5%
α_C (cycling effect)	0.152	0.148	−2.5%
γ (rebound)	1.669	1.408	−15.7%
Random effect SD (σ_{track})	—	0.59	—

Table 3: Comparison of key parameters between fixed-effects NB-GLM and NB-GLMM. The kernel amplitude effects (α, α_I, α_C) differ by less than 3.5%, confirming that the main findings are robust to inclusion of random intercepts. The random effect SD of 0.59 indicates substantial between-track variation in baseline event rate.

Table S3: Condition-Specific Suppression Amplitudes

The kernel amplitude varies across conditions while maintaining invariant shape and timescales:

Condition	Amplitude	Events	Tracks	Interpretation
0→250 | Control	1.005	1,407	99	Reference condition
0→250 | Cycling	1.157	2,410	214	+15% (cycling enhancement)
50→250 | Control	0.340	2,440	187	−66% (partial adaptation)
50→250 | Cycling	0.492	1,031	123	Combined effects

Table 4: Suppression amplitudes computed as $\alpha + \alpha_I \cdot I + \alpha_C \cdot C$.

Figure S1: Residual Diagnostics

Figure S2: Time-Rescaling Test

The time-rescaling test assesses whether the fitted hazard model produces inter-event intervals consistent with a Poisson process. Under the correct model, rescaled inter-event times should follow Exp(1), and their cumulative distribution should be uniform.

Diagnostic Statistics

- **Pearson residuals:** Mean = 0.0001, SD = 1.01, Skew = 40.7, Kurtosis = 2440
- **Large residuals:** 7,288 observations (0.09%) with $|r| > 3$
- **Time-rescaling:** KS statistic = 0.041, p = 0.17, mean deviation = 1.3%

The high skewness and kurtosis of residuals reflect the zero-inflated nature of event data (most frames have no events). The time-rescaling test passes at conventional significance levels, supporting the adequacy of the hazard model specification.

Software and Data Availability

All analyses were performed in Python 3.14 using:

- statsmodels 0.14 (GLM fitting)
- numpy 2.0, pandas 2.2 (data manipulation)
- matplotlib 3.9 (visualization)
- Quarto 1.4 (document preparation)

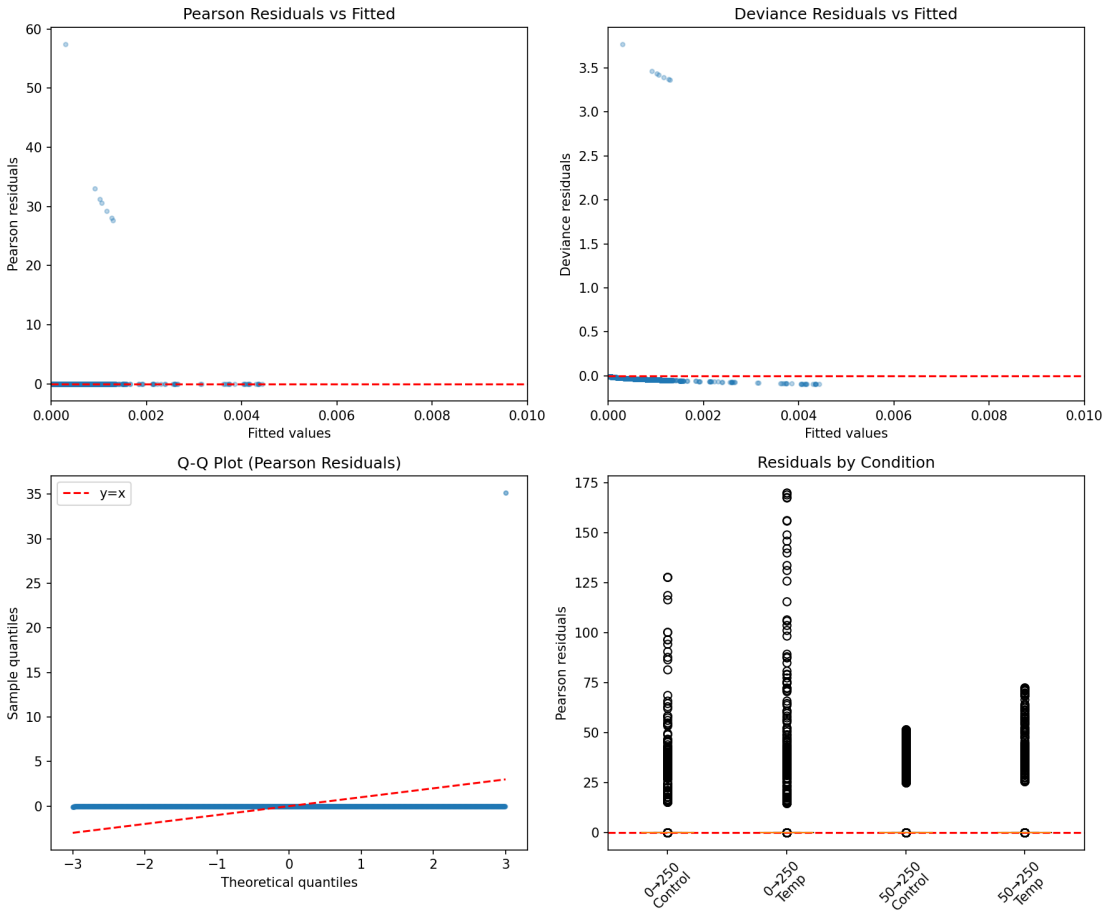

Figure 1: **Factorial model diagnostics.** (A) Pearson residuals vs fitted values show no systematic pattern. (B) Deviance residuals vs fitted values. (C) Q-Q plot of Pearson residuals against theoretical normal quantiles. (D) Residual distributions by condition show similar spread across all four experimental conditions. Residual mean = 0.0001, SD = 1.01.

Code and processed data are available at: https://github.com/GilRaitses/INDYsim

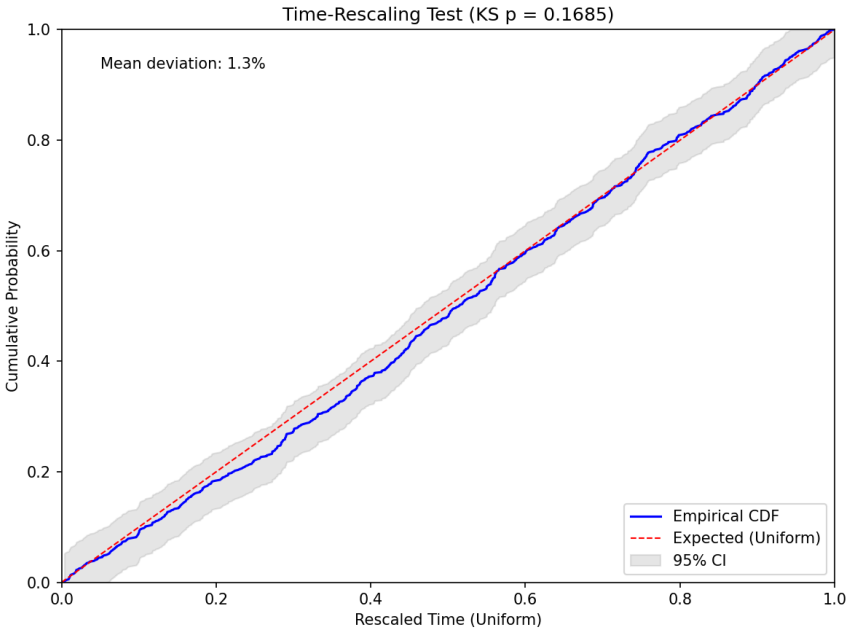

Figure 2: **Time-rescaling test.** Empirical cumulative distribution of rescaled inter-event times (blue) compared to expected uniform distribution (red dashed). Gray shading indicates 95% confidence band. KS test: $D = 0.041$, $p = 0.17$. Mean deviation = 1.3%, indicating adequate model fit.